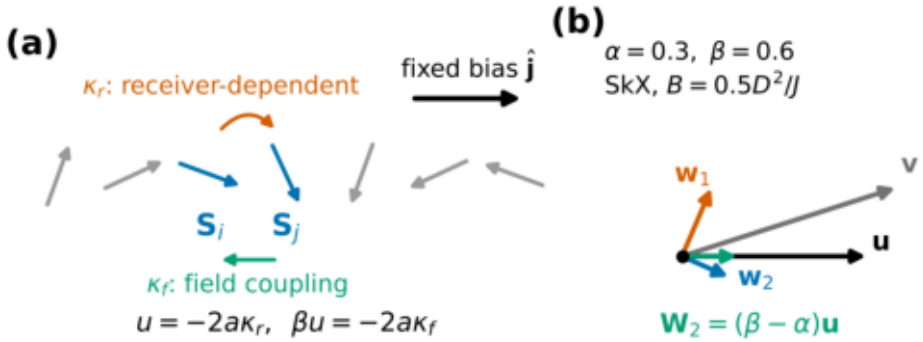


Fig. 1

A current is a fixed nonreciprocal bond bias, and in the co-moving frame it enters only through two vectors of opposite time-reversal parity. (a) On a bond i → j the adiabatic term is a receiver-dependent coupling κ_r and the β term a field coupling κ_f, with u = −2aκ_r and βu = −2aκ_f. (b) Slot vectors of the driven skyrmion lattice (B = 0.5D²/J, α = 0.3, β = 0.6) from the exact Thiele projection. $w_1 = v - u$ (T-odd) and $w_2 = \beta u - \alpha v$ (T-even) are perpendicular, and the Landau–Lifshitz drive $W_2 = w_2 + \alpha w_1 = (\beta - \alpha)u$ points along u whatever the Hall angle of v.

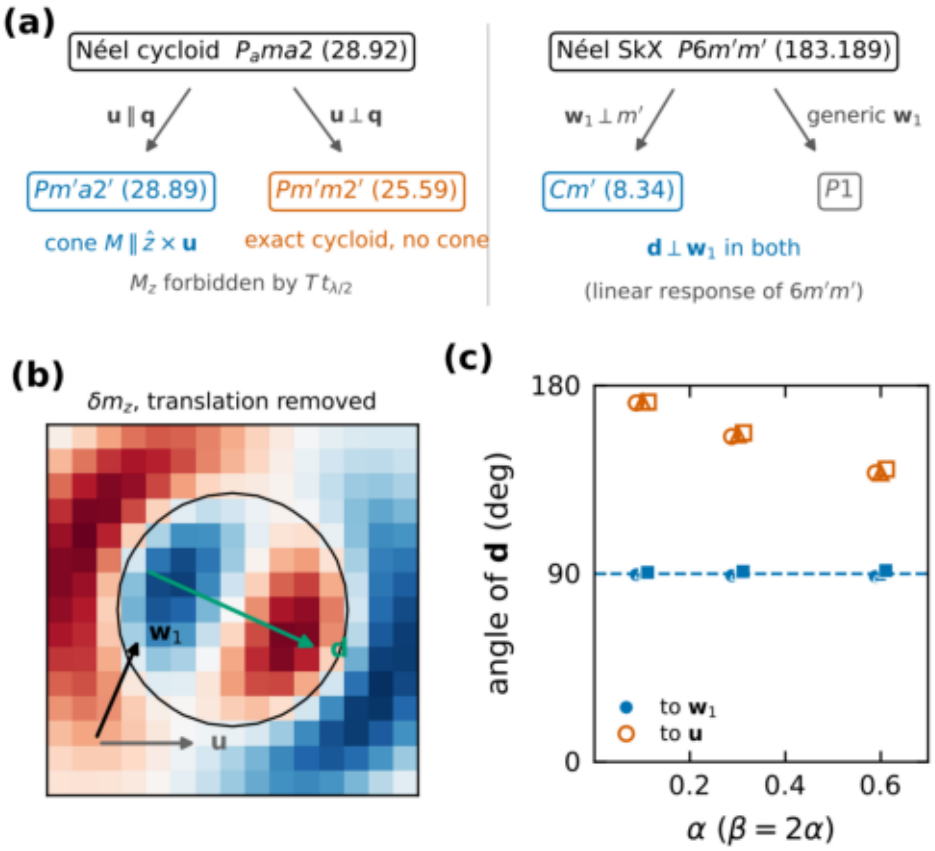


Fig. 2

The residual magnetic group of the driven pattern fixes its first-order response, and the vector that matters is w₁, not u. (a) Group–subgroup steps with BNS numbers (spglib on minimal commensurate cells). Cycloid: u ∥ q leaves Pm′a2′ and allows a uniform cone M ∥ ẑ×u; u ⊥ q leaves Pm′m2′ and the cycloid stays exact; the out-of-plane cone is forbidden by the half-period antitranslation T t_{λ/2}. Skyrmion lattice: linear response of 6m′m′ gives a dipolar distortion d = a ẑ×w₁ + b w₂, perpendicular to w₁. (b) Change of m_z of a driven Néel skyrmion (α = 0.3, u = 0.02 along x) after the uniform translation is removed. Black contour: m_z = 0 of the static skyrmion. Green arrow: first moment d inside r < 5. (c) Angle of d measured from w₁ (filled) and from u (open) for α = 0.1, 0.3, 0.6 and current directions 0°, 30°, 90° (circle, triangle, square). d stays perpendicular to w₁ within 1.5°; its angle to u follows the Hall rotation.

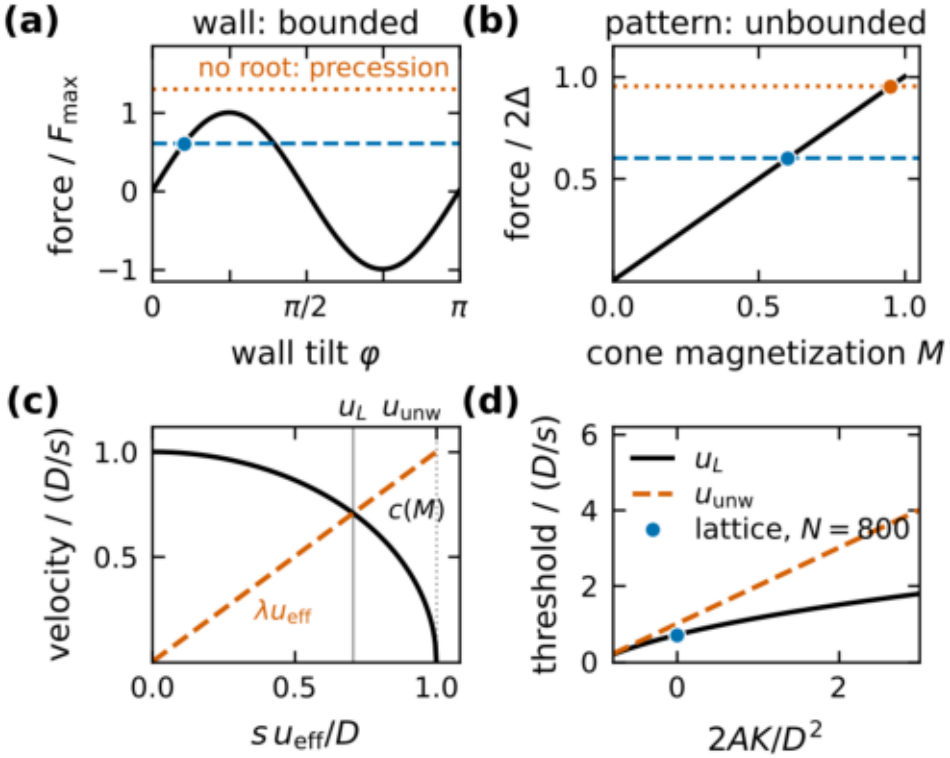


Fig. 3

A pattern has no precessional Walker breakdown; its static branch ends in a Landau instability before it can unwind. (a) The conjugate variable of a domain wall, the tilt φ, has a bounded restoring force. A drive above F_max (dotted) has no static root and the wall precesses. (b) The conjugate variable of a pattern, the cone M, has a quadratic energy, so a static root exists up to |M| = 1. (c) Helix with exchange and DMI: the compression velocity on the driven cone, c = (D/s)√(1 − M²) with M = su_eff/D, meets the Doppler line λu_eff (λ = 1) at u_L = D/(√2 s), below the unwinding point u_unw = D/s. (d) With easy-axis anisotropy K, u_L = Λ/√(1 + κ²) stays below u_unw = Λ/κ for every K (Λ = √(D² + 2AK)/s, κ = D/√(D² + 2AK)). Point: exact Bloch threshold of an N = 800 chain.

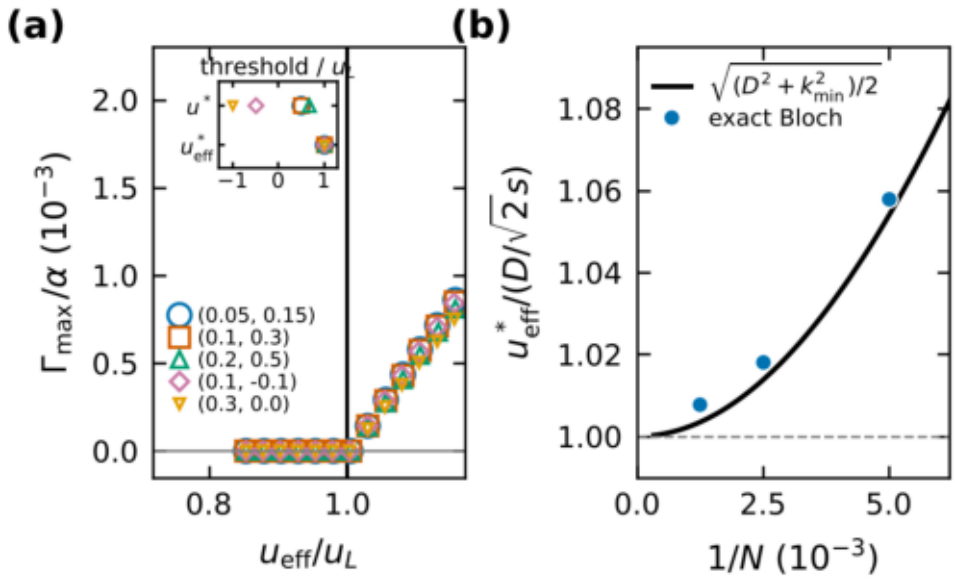


Fig. 4

The driven helix breaks at u_L = D/(√2 s) for every (α, β): the threshold depends on β only through u_eff = (β/α − 1)u. (a) Largest Bloch growth rate divided by α against u_eff for five (α, β), including β = 0 and β < 0 (N = 200; u_L includes the finite-chain correction √((D² + k_min²)/2)). Inset: the same thresholds in the bare current u (spread over a factor of two and both signs) and in u_eff (equal to 0.1%). (b) Finite chains: exact thresholds (points) against √((D² + k_min²)/2) (line), converging to D/√2 (dashed).

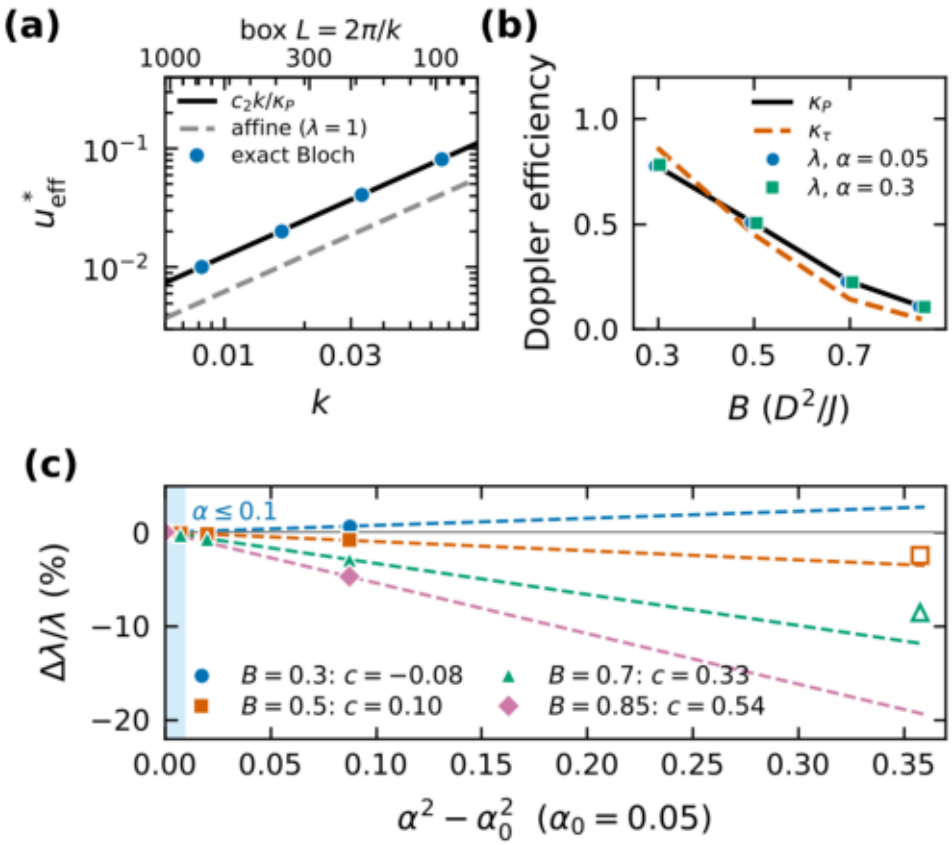


Fig. 5

The skyrmion-lattice threshold is u_eff* = c₂k/λ with λ = κ_P, the dissipative overlap; damping enters only at order α². (a) Exact Bloch thresholds (B = 0.5D²/J, α = 0.3, β = 0.6) at k = 2π/(19n), n = 40, 20, 10, 5, against c₂k/κ_P with c₂ and κ_P from the static lattice (line; agreement within 0.4–0.9%) and the affine estimate λ = 1 (dashed). Top axis: box length L = 2π/k, so the threshold falls as 1/L. (b) λ from exact thresholds at α = 0.05 and 0.3 lies on κ_P (solid), not on κ_τ (dashed), at four fields. (c) Relative change of λ from α₀ = 0.05, Δλ/λ = −c(α² − α₀²), fitted for α ≤ 0.3 (filled, dashed lines). Open symbols: α = 0.6, near critical damping. Shaded band: α ≤ 0.1, where λ = κ_P within 0.6%.

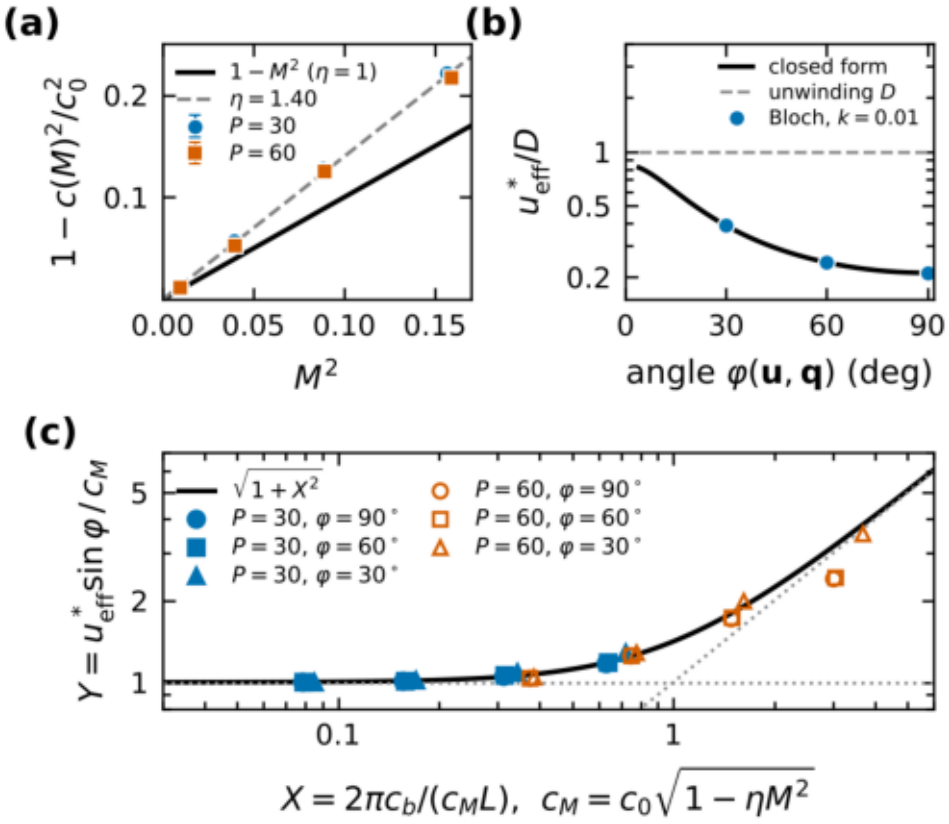


Fig. 6

Stripes have a quadratic Goldstone mode: a current component along q only scales the bending velocity, and every finite-box threshold collapses onto u_eff* sin φ = √(c_M² + (2πc_b/L)²). (a) k-resolved bending velocity on converged driven states (u ∥ q, residual < 10^{-12}; error bars: spread over k = 0.0025–0.04 q). $1 - c(M)^2/c_0^2 = \eta M^2$ with η ≈ 1.4, the same for every k and for P = 30 and 60, steeper than the exchange value η = 1 (solid). (b) Threshold against the current angle (P = 30): exact Bloch thresholds at k = 0.01 (points) and the k → 0 closed form $u^* = c_0/\sqrt{(\sin^2\varphi + \eta(c_0/D)^2\cos^2\varphi)}$ (line). (c) Dynamical thresholds in sheared boxes of length L = 64–512 (request 6) for P = 30 (filled) and 60 (open) and φ = 90°, 60°, 30° collapse onto $Y = \sqrt{(1 + X^2)}$, with $c_M = c_0\sqrt{(1 - \eta M^2)}$. The P = 60, L = 64 points leave the curve where the box wavenumber approaches the stripe wavenumber.

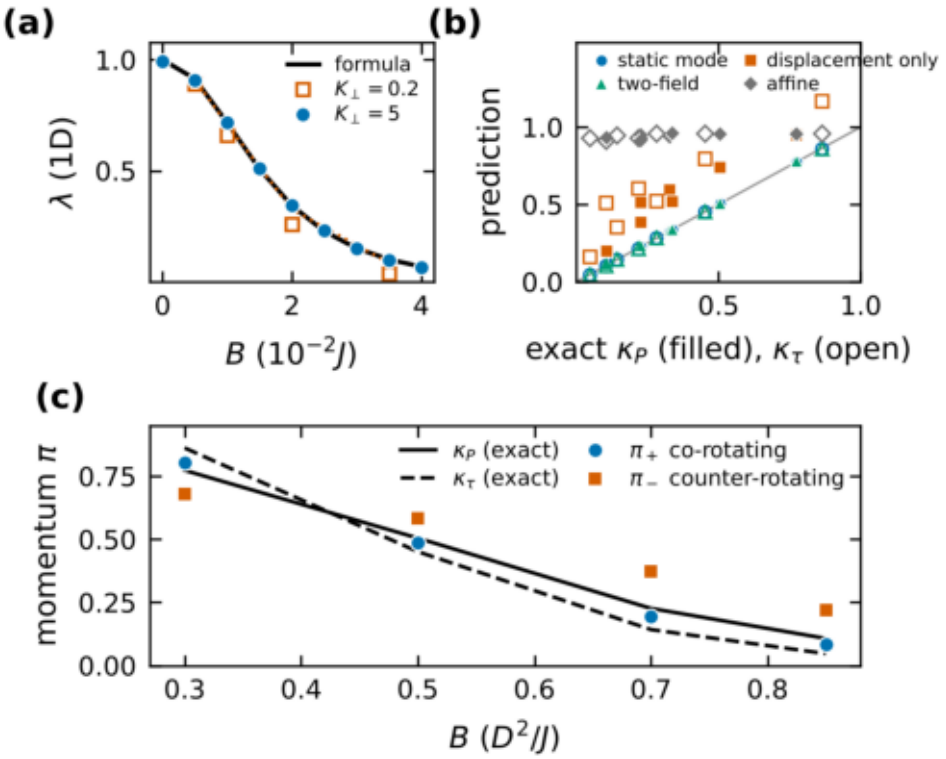


Fig. 7

λ is a static property of the pattern: one minus the counterflow momentum carried by its internal relaxation. (a) 1D chiral soliton lattice (D = 0.3, period 40): exact Doppler overlap against the homogenization formula $\langle 1/M\rangle/(\langle\psi_0^{-2}\rangle\langle\psi_0^2/M\rangle)$ with a slaved conjugate field ($K_\perp = 5$). With $K_\perp = 0.2$ the conjugate field is not slaved and the formula (dotted) overestimates λ. (b) Seven skyrmion lattices: predictions against the exact dynamic overlaps κ_P (filled) and κ_τ (open). The static relaxed mode and the two-field Ritz theory (displacement plus canting, no fitted parameter) fall on the diagonal; displacement-only theory and the affine mode do not. (c) Circular decomposition of the Bloch mode (19×33): the counterflow cancels mainly the co-rotating momentum π_+; the counter-rotating π_- keeps more, so $\kappa_P > \kappa_\tau$ once $\pi_- > \pi_+$.

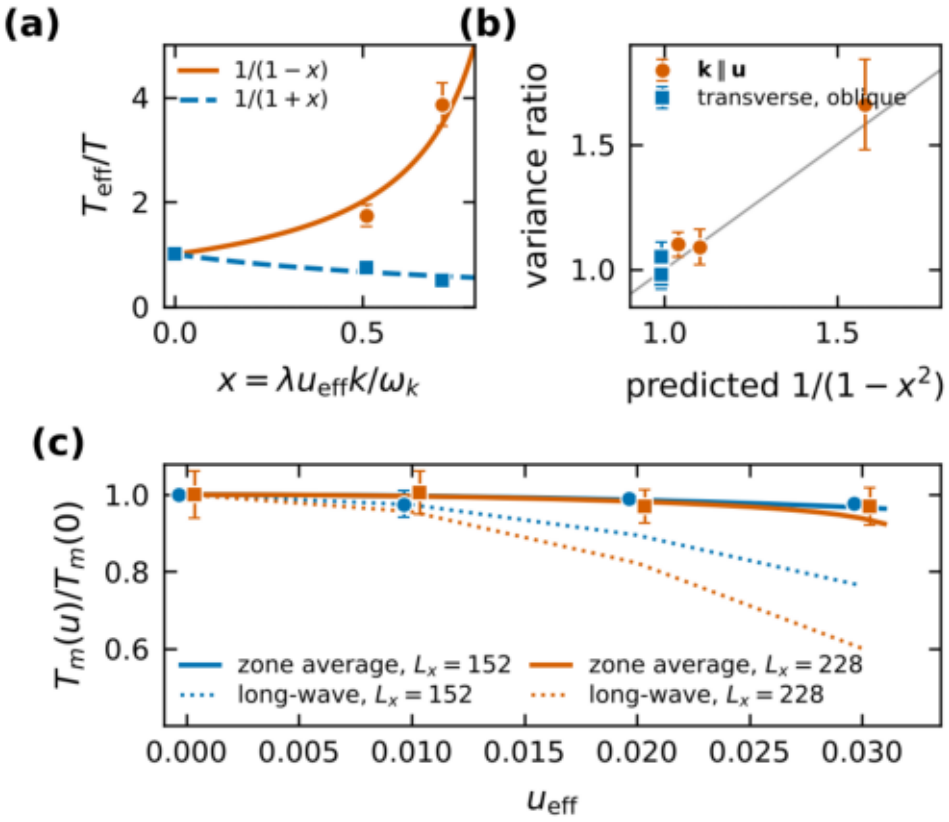


Fig. 8

Below threshold the current puts the lattice phonons in a Gibbs ensemble moving at λu_eff, and the melting point moves only by the zone average of that heating. (a) Effective temperatures of the two branches at $k_1 = 2\pi/76$ (stochastic LLG, 76×33, k_BT = 0.01) against $1/(1 \mp x)$. (b) Variance ratio of displacement modes in a 152×66 box at u_eff = 0.03 against $1/(1 - x^2)$. Modes along the current heat; transverse and oblique modes stay at the bath temperature. (c) Melting temperature from heating ladders (256–264 skyrmions, three seeds) against the zone average $1/\langle 1/(1 - x^2)\rangle_{BZ}$ (solid) and the rejected long-wave criterion (four longest modes, dotted), for two box shapes.

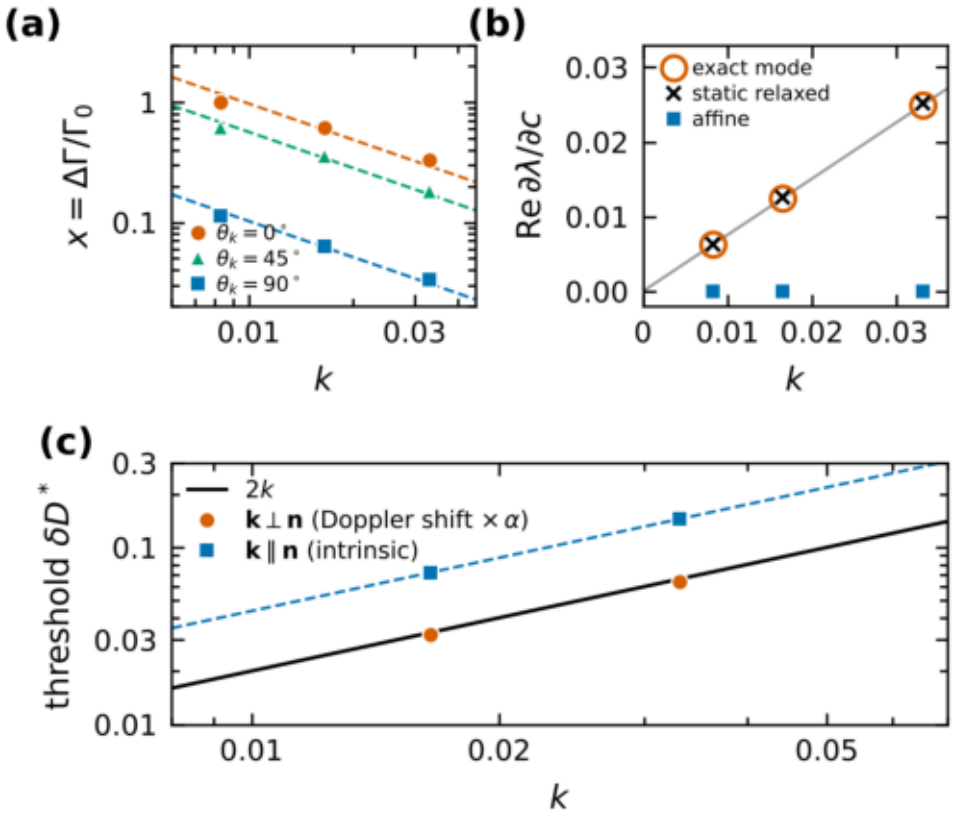


Fig. 9

Exchange-even couplings pump the lattice phonons with zero threshold through the static relaxed mode, the dual of the Zhang–Li Doppler term. (a) Damping-like SOT (c = 0.004, n = ŷ, α = 0.3): pumping ratio x = ΔΓ/Γ₀ against k for three directions of k. x·k is nearly constant (dashed: 1/k), so the threshold vanishes in an infinite system. (b) First-order generator Re ∂λ/∂c on the exact mode, the static relaxed mode and the affine mode (α = 0). The relaxed mode carries all the pumping; the affine mode carries none. (c) Second-order nonreciprocal stiffness of δD (n = ŷ, α = 0.3): threshold δD* = Γ₀/(half-split per δD) for k ⊥ n (Doppler shift converted by damping) and k ∥ n (intrinsic splitting). Both are ∝ k.

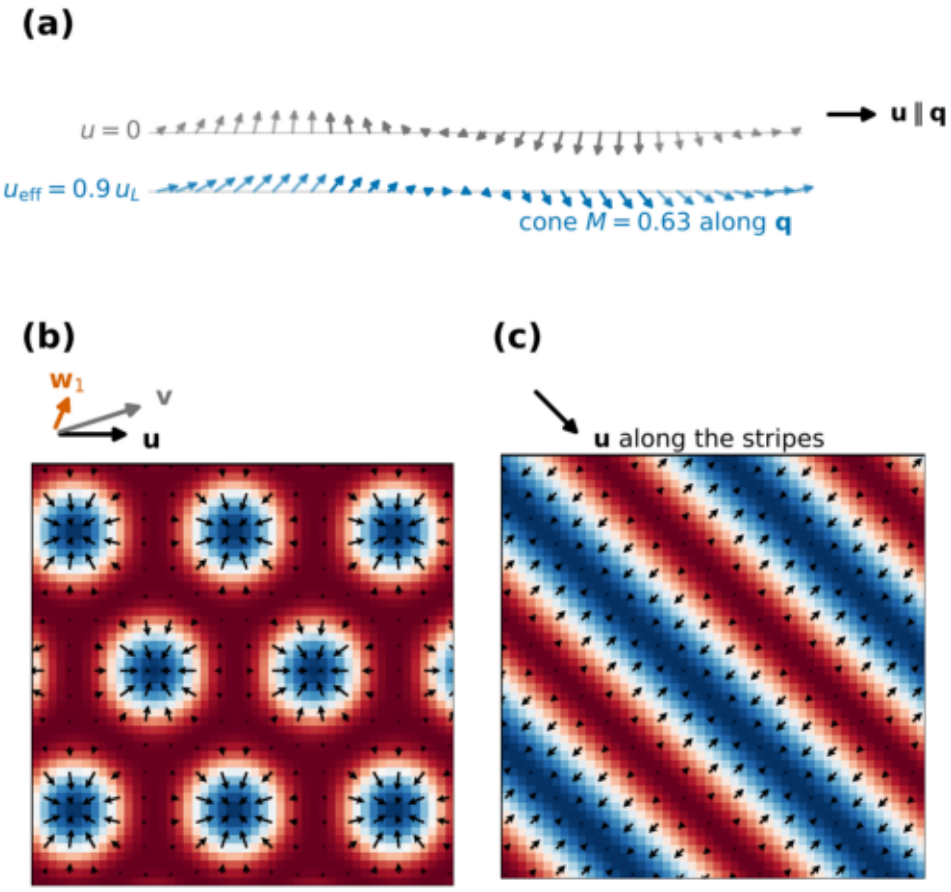


Fig. P1

The three driven chiral patterns, and what a current does to each to first order. (a) Bloch helix (exact driven cone of `code/helix`, every second spin of one period, oblique view): unbiased, and at u_eff = 0.9 u_L, where the spins tilt into a cone M = s u_eff/D along q. (b) Néel skyrmion lattice (B = 0.5D²/J; colour m_z from −1, blue, to +1, red; arrows in-plane spin): current u, Hall velocity v and the slot vector $w_1 = v - u$ from the exact Thiele projection (α = 0.3, β = 0.6). (c) Diagonal Néel stripes (P = 30, converged state) with a current along the stripes, the direction that couples to the bending Goldstone mode.

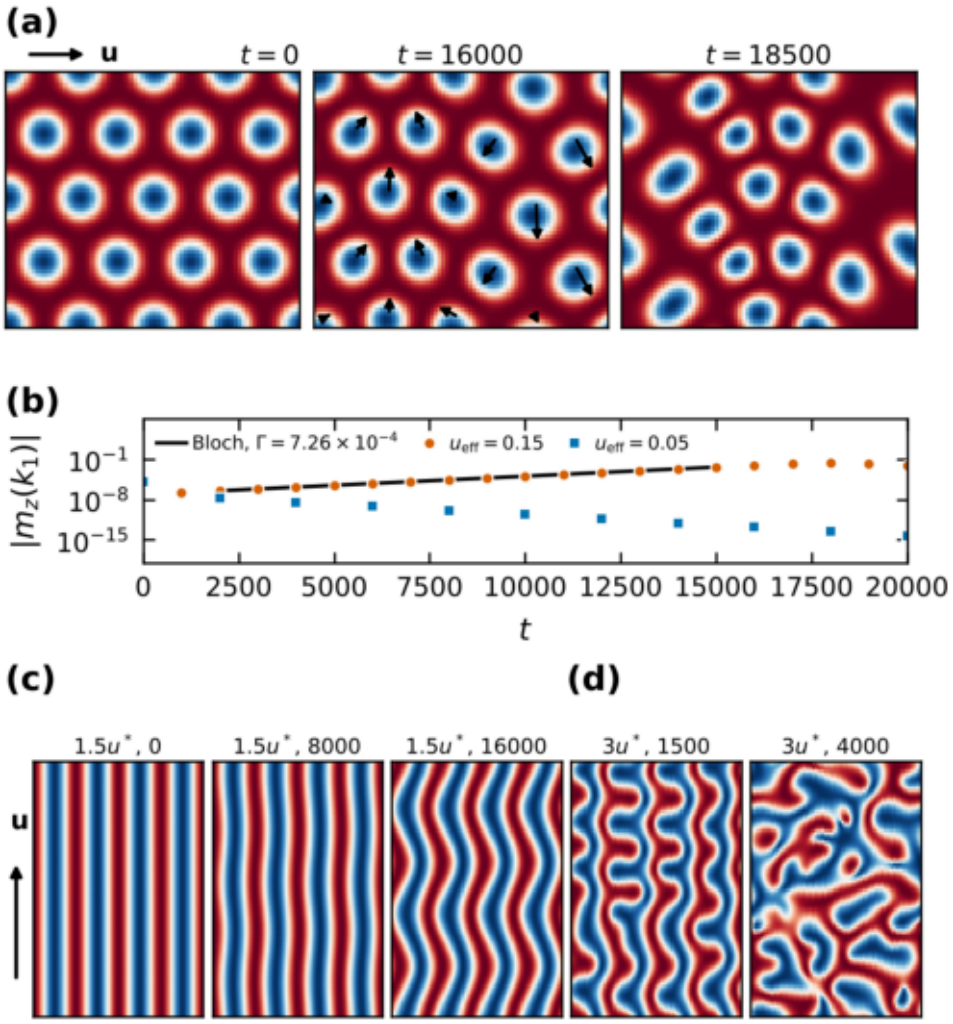


Fig. P2

Above the Doppler threshold the Goldstone mode of the pattern grows at the rate of the Bloch analysis and then reorganizes the pattern. (a) Skyrmion lattice at u_eff = 0.15 (1.5 times the threshold of the $k_1 = 2\pi/76$ mode; 76×66 box, 16 skyrmions, co-moving frame, noise 10^{-3} at t = 0). Arrows at t = 16000: skyrmion displacements since t = 0 (×4), a box-length wave along the current; at t = 18500 the lattice is strongly disordered (the flowing state is treated in paper B2). (b) Satellite amplitude $|m_z(k_1)|$ against time: dynamics (points) follows the Bloch growth rate of the same mode on the driven stationary state (line, $\Gamma = 7.26\times10^{-4}$, no fit of the slope); below threshold (u_eff = 0.05) the noise decays. (c) Stripes (P = 30, box 85 × 256 across × along, sheared periodic box) driven along themselves at 1.5u*: the longest bending wave grows and saturates into a zigzag. (d) At 3u*: shorter bending waves, then dislocations and a disordered labyrinth. Panels (c, d) show m_z in the frame of the stripes.

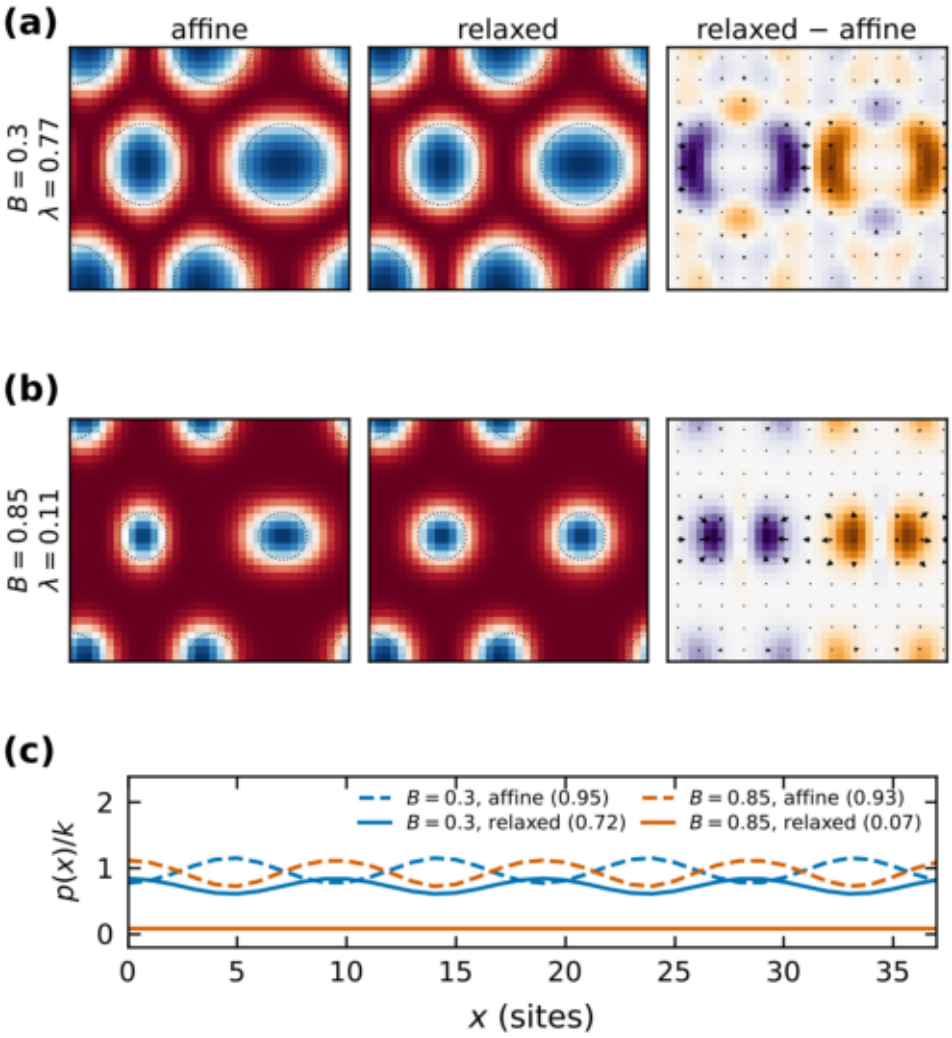


Fig. P3

λ in real space: under a long-wave strain the skyrmions keep their shape, and the internal canting that does this carries a counterflow that cancels part of the Doppler momentum. (a, b) A longitudinal strain wave (wavelength two cells, displacement along x of amplitude 2 sites, exaggerated) on the lattice at B = 0.3D²/J (λ = 0.77) and 0.85D²/J (λ = 0.11). Left: affine mode, in which the skyrmions are squeezed and stretched with the lattice. Middle: static relaxed mode with the same skyrmion displacement, in which the skyrmions stay round. Right: their difference, the internal canting (colour δm_z, arrows in-plane). Dotted: m_z = 0 of the undeformed lattice. (c) Doppler-momentum density of the phonon (each mode with its own elliptic polarization), averaged across the strip and divided by k; the averages (in brackets) are the Doppler overlaps of the two modes at this wavelength. The affine mode carries nearly the full momentum at both fields; the relaxation removes a quarter of it in the dense lattice and almost all of it in the dilute one.